

Improving Motor Imagery Brain Computer Interface Classification

Group 1

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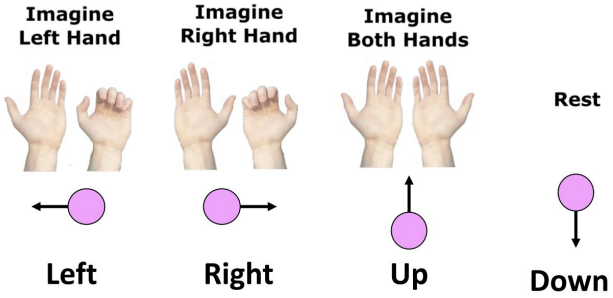
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Introduction

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Experiment Background

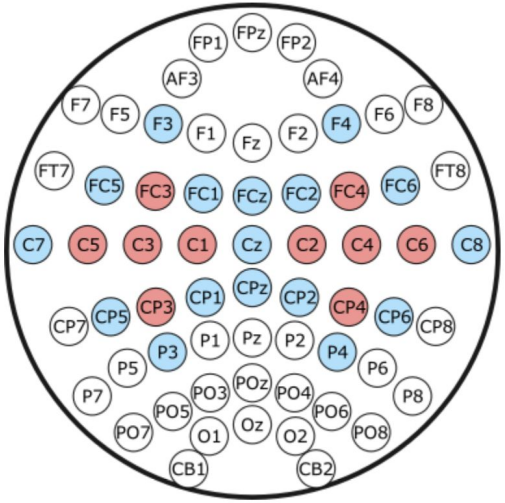


Motor imagery (MI) task:

- Participant *imagines* moving left vs. right hand, and both (up) vs. none (down) in the other experiment, without actual movement.
- MI modulates the **sensorimotor rhythm (SMR)**:
  – **Mu (8–13 Hz)** and **beta (13–30 Hz)** over the sensorimotor cortex.
  – Imagining right-hand movement → **ERD (power decrease)** over **C3**;
  imagining left-hand movement → ERD over **C4**.

Electrode montage:

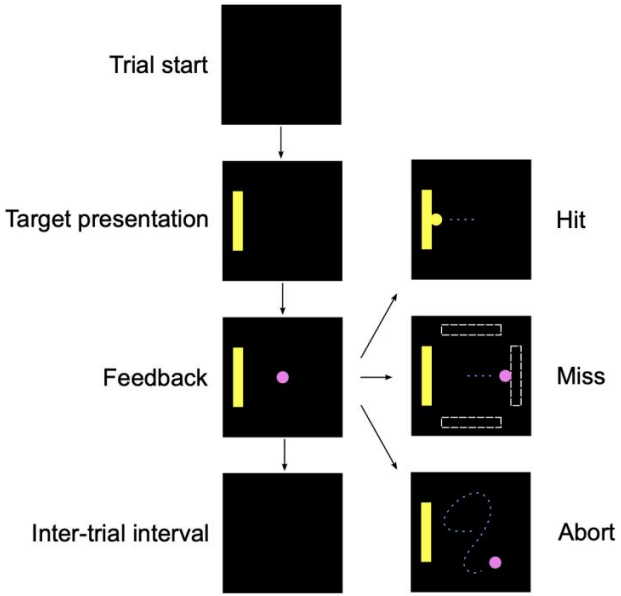
- 64-channel cap; analysis focuses on **sensorimotor channels**.
- **Online BCI (red electrodes):** small set around C3 and C4 used in real time.
- **Offline analysis (red + blue):** extended set of ~27 gelled channels over FC–C–CP rows to capture more spatial information.



“Motor imagery changes alpha/mu power over contralateral sensorimotor cortex; we record from C3/C4-centered electrodes to capture these SMR changes.”

PVC vs PTC and Online Classifier

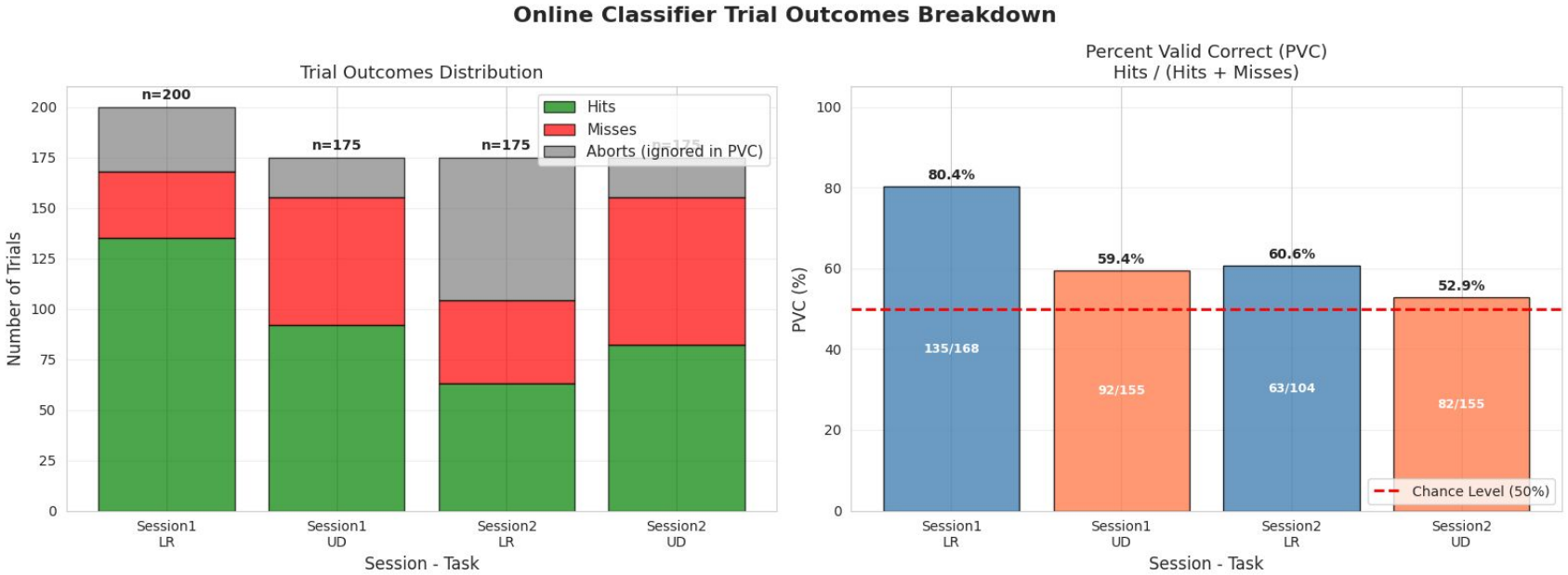
$$PVC = \frac{\#Hits}{\#Hits+\#Misses} \qquad PTC = \frac{\#Hits}{\#Hits + \#Misses + \#Aborts}$$



- **PTC (Percent Total Correct):** overall behavioral score – % of all trials where the final cursor/ball ends on the correct target (aborts counted as incorrect).
- **PVC (Percent Valid Correct):** classifier-centric score – % of **hit vs. miss** trials where the online classifier sends the cursor in the correct direction (aborted trials ignored).
- **Online linear classifier:** simple FT+LDA (using Burg AR) to estimate C3/C4 alpha power and their difference, then a 1-D LDA decides left vs. right (or up vs. down).
- **Why we use PVC:** our offline models are also evaluated on **non-aborted trials only**, so using PVC (and skipping aborted trials) makes our performance directly comparable to the original online linear classifier.

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Online Performance



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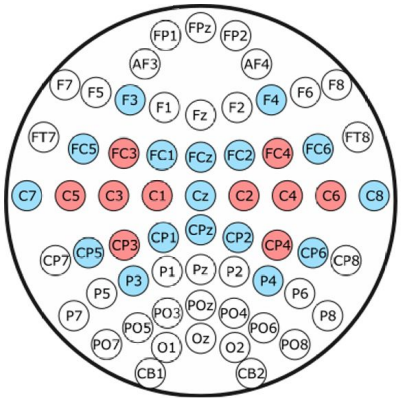


Preprocessing + Feature Selection

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Preprocessing

Data → **Selecting Motor ROI** → **Bandpass** (~8-30 Hz) → **CAR** → **Down-sampling** → Epochs

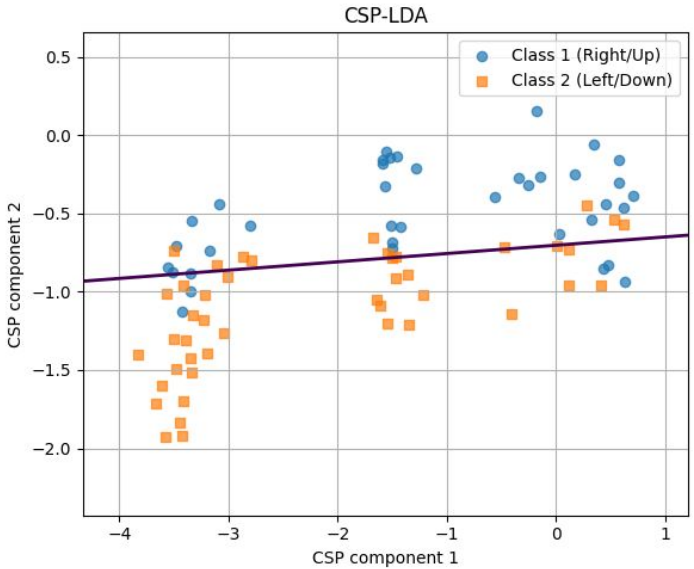


Optional:
- **Laplacian**
- **ICA**
- **Epoch exclusion criteria (too short, trial aborted)**

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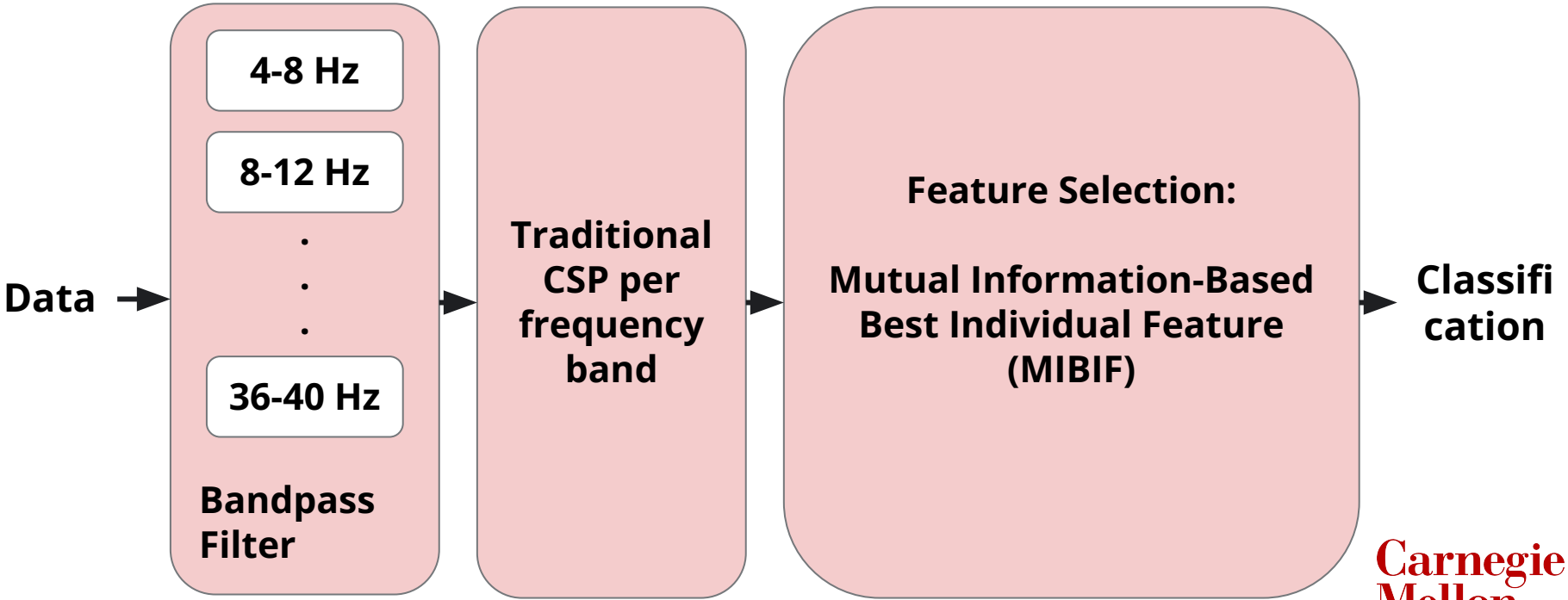
Common Spatial Pattern

- CSP identifies spatial filters with **maximal variance for one class and minimal variance for the other**, enhancing discriminative features.
    - Motor imagery produces **distinct spatial patterns** in **mu (8–13 Hz)** and **beta (13–30 Hz)** rhythms over sensorimotor cortex.
- The EEG is often **band-pass filtered into multiple frequency bands** (e.g., filter bank CSP) to capture band-specific discriminative features.



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Filter Bank CSP (FBCSP)



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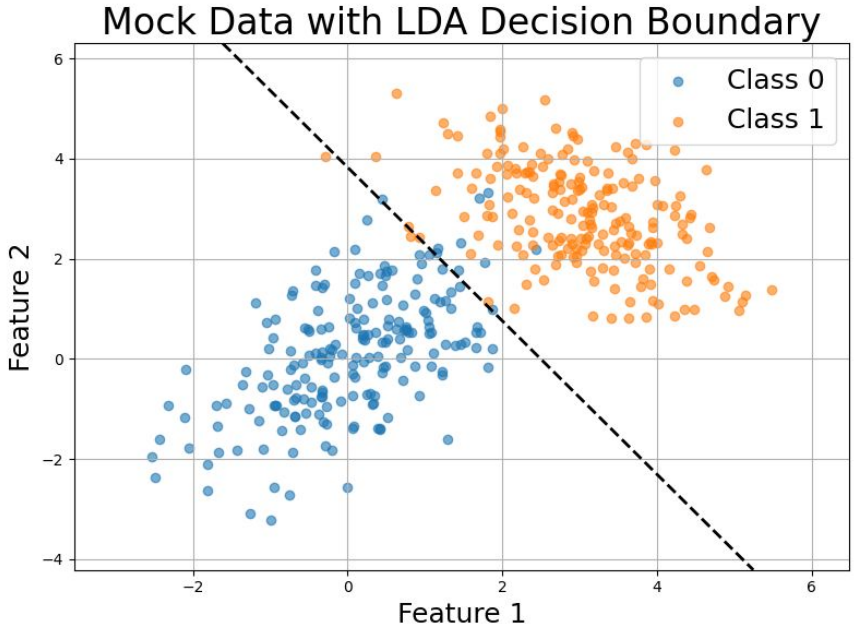
Classifiers

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Classifiers

1. **Linear Discriminant Analysis (LDA)**
   - Computes the different means and shared covariance of given classes,
   - and then finds the projection that maximizes the separation of the means, but minimizes the within-class variance.

2. **Shrinkage LDA (sLDA)**
   - If the number of features is large, but the number of samples is low, then the covariance matrix is unstable.
   - Replace covariance matrix in LDA with $\Sigma_\lambda = (1-\lambda)\Sigma + \lambda I \rightarrow$ Make it more spherical.



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Classifiers

3. **Support Vector Machine (SVM)**
    - Maximize distance between decision boundary and closest points of each class.
    - Better when the features are not Gaussian or have equal covariance.

→ Generally, these classifiers all perform at around the same level.

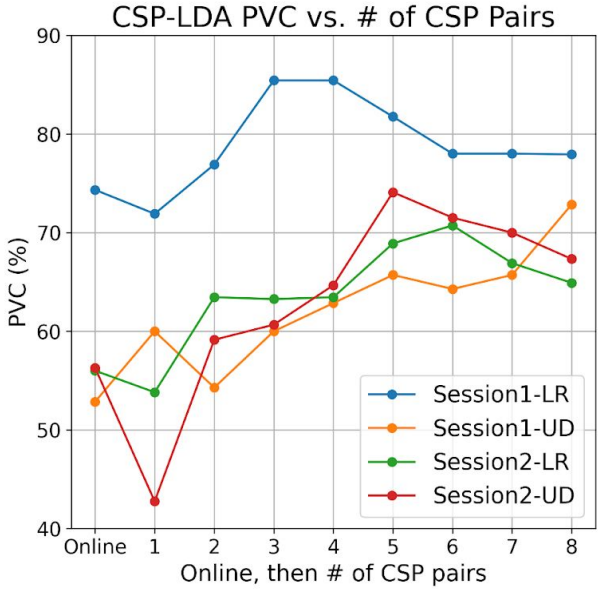
4. **Random Forest (RF)**
    - Random Forest is an ensemble method that trains many decision trees on bootstrapped subsets of the data and uses random subsets of features at each split. Each tree votes on the class label, and the final prediction is the majority vote. RF can model complex non-linear decision boundaries, is relatively robust to noise and overfitting, and works well when there are many input features without strong assumptions on their distribution.

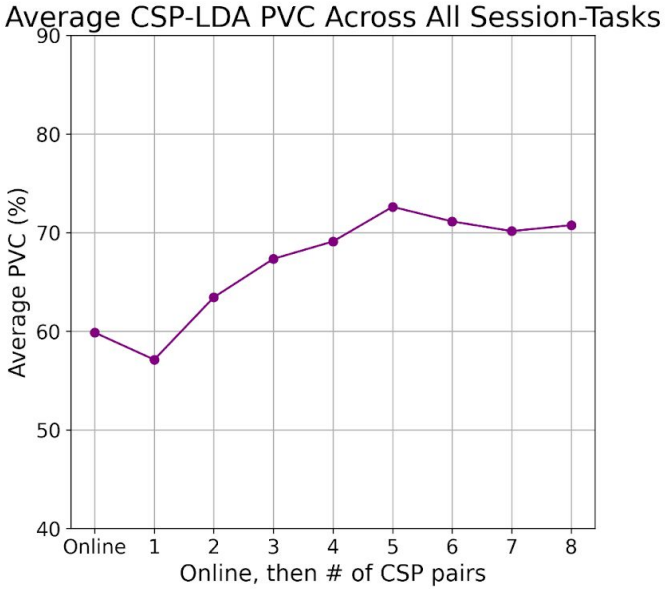


CSP/FBCSP Results

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Exploring Impact of # of CSP Pairs



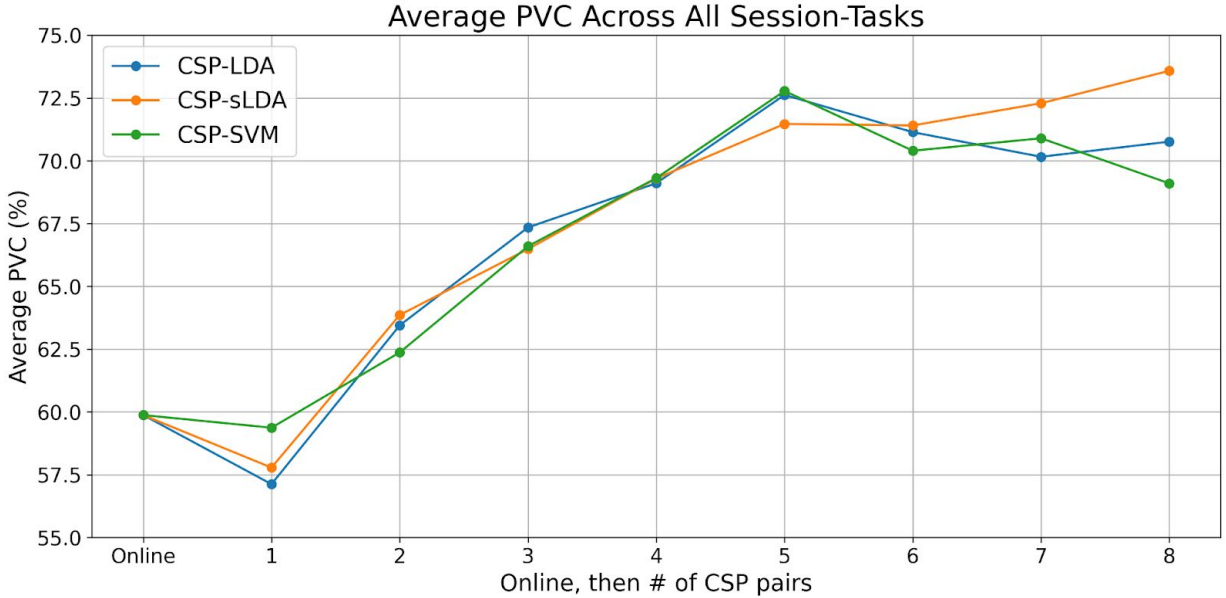


5 pairs of CSP features is **optimal.**

Behavior is consistent between session-tasks.

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Comparing Different Classifiers



SVM performs the best at 5 CSP pairs.

sLDA seems to overfit as PVC continues to climb unexpectedly.

LDA or SVM are most suitable for CSP.

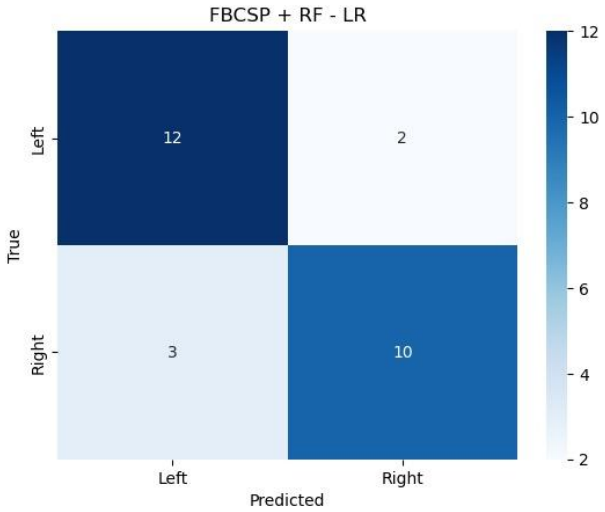
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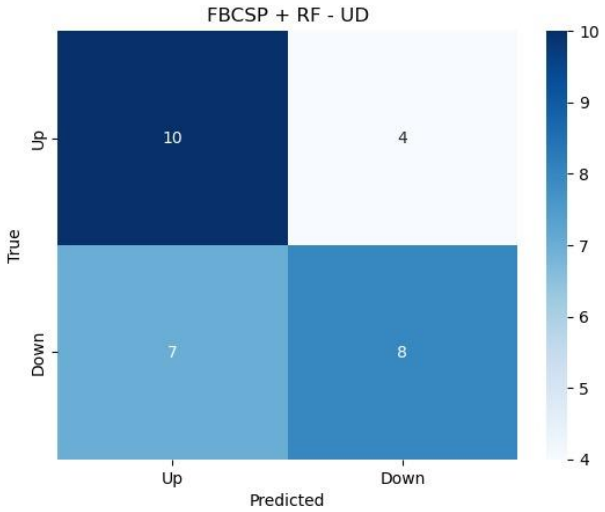
Applying FBCSP Features to a RF Model

Separating CSP features by frequency bands per task gives us higher PVC for a single task:

- LR task: 81.5%
- UD task: 62.1%

UD task suffers as it evenly classifies down as up.





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Deep Learning

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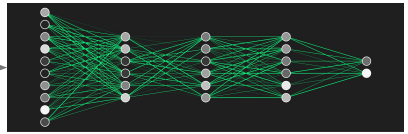
Neural Networks and BCI

- Neural networks are black boxes, not very interpretable
- Can make their own features from the data. They can also classify based on existing features
- Computational cost of neural network architectures is high, especially during training
- Subject specificity in deep learning often dominates performance
- When we rejected all trials that were smaller than 1500 samples, this resulted in only keeping 329 valid trials total. This is NOT a lot of data to work with in a deep learning setting
    - This lead to high regularization required if learning from EEG data directly
- Deep learning achieves state of the art performance

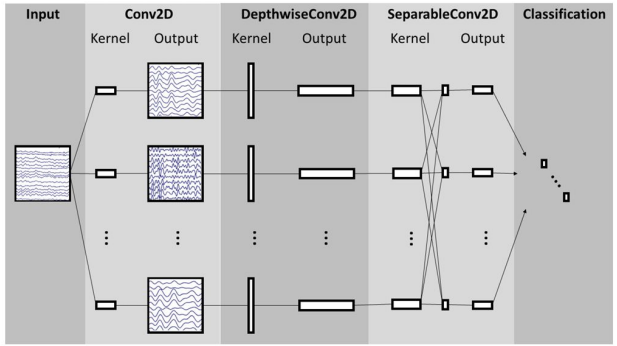
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Neural Network

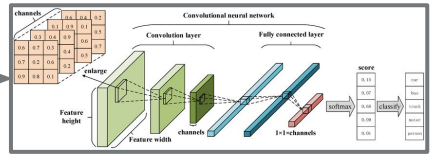
- There a huge variety in flavor of neural networks

1) CSP → 

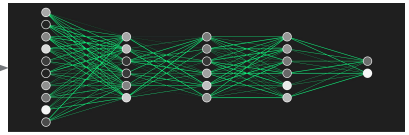
MLP

2) raw EEG data → 

EEGNet (Lawhern et al., 2018)

3) CNN: CSP → 

CNN

4) CSP + raw EEG data → 

Feature-fusion network

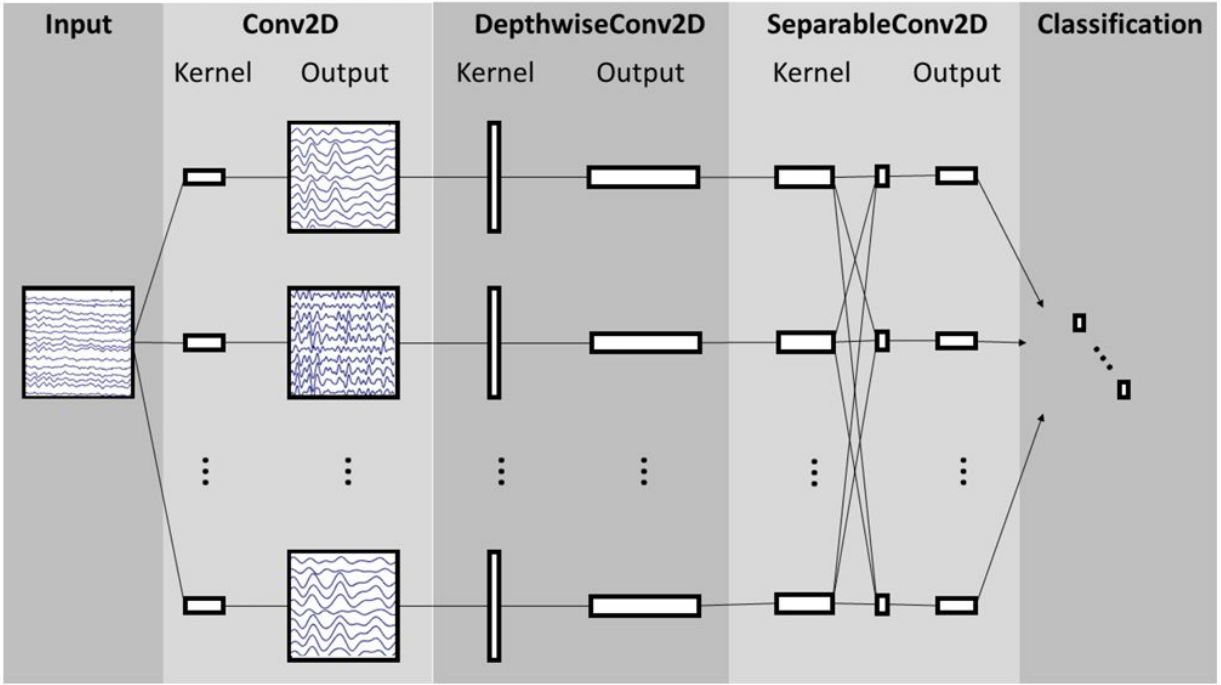
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EEGNet

EEGNet is built analogously to EEG processing

- input: raw EEG data
- requirements: preprocessed signal

1. Temporal Convolution→ Band Pass Filter
2. Depthwise Convolution → Spatial Filtering (CSP-like)
3. Classification → Simple Classifier model (e.g. LDA)



(Lawhern et al., 2018)

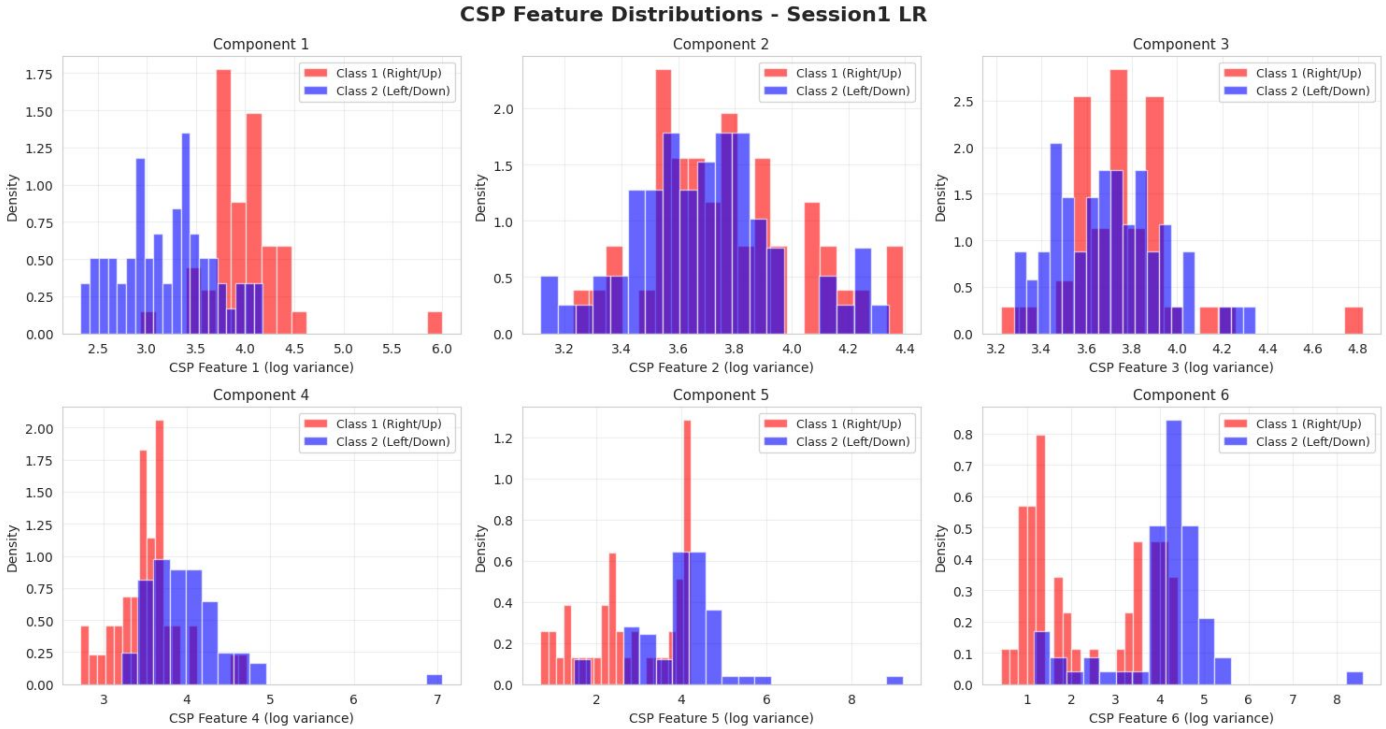
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Deep Learning Results

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Subject 1 (LR Task)



Histograms show the log-variance of six CSP components for Class 1 (Right) and Class 2 (Left).

Clear separation in several components indicates effective spatial filtering and discriminative power between motor-imagery classes.

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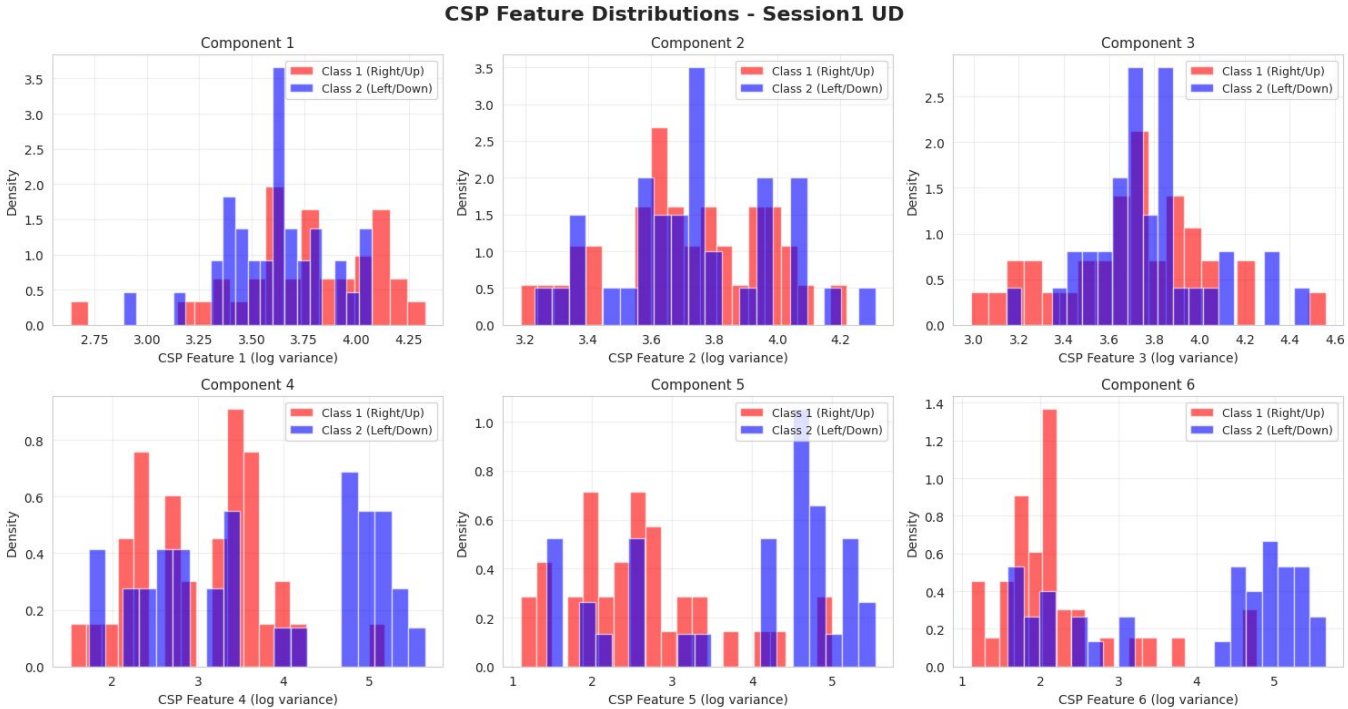
Subject 1 (LR Task)



The left panel shows training and test loss over 50 epochs, indicating stable convergence but some oscillation in test loss. The right panel shows accuracy curves, with training accuracy reaching 100% and test accuracy stabilizing around ~85%, suggesting mild overfitting but good generalization overall.

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Subject 1 (UD Task)

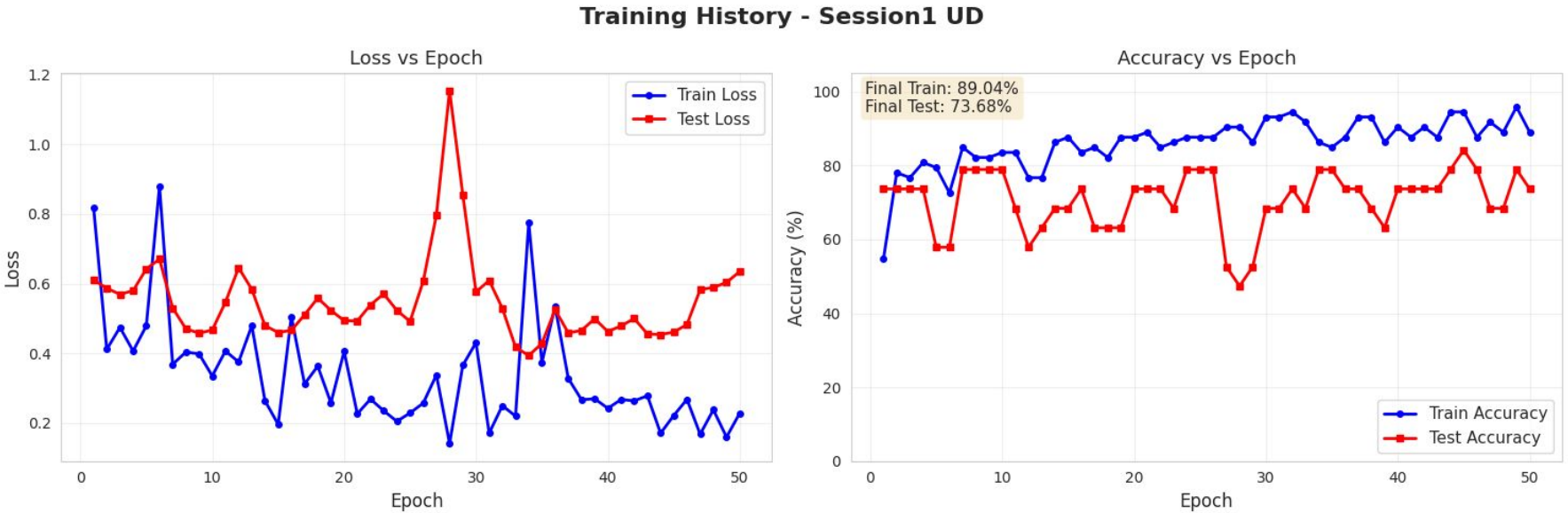


Histograms show the log-variance of six CSP components for Class 1 (Up) and Class 2 (Down).

Clear separation in several components indicates effective spatial filtering and discriminative power between motor-imagery classes.

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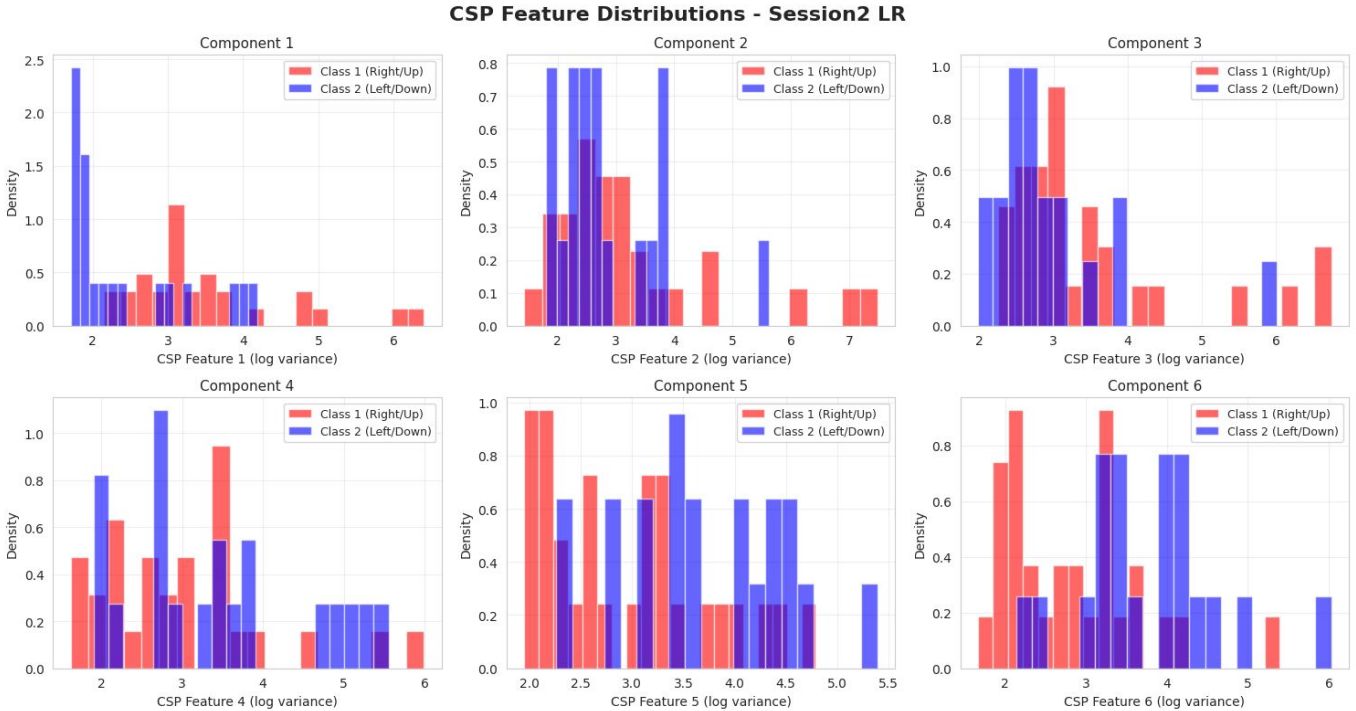
Subject 1 (UD Task)



The left panel shows training and test loss over 50 epochs, indicating stable convergence but some oscillation in test loss. The right panel shows accuracy curves, with training accuracy reaching 89% and test accuracy stabilizing around ~73%

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Subject 2 (LR Task)



Histograms show the log-variance of six CSP components for Class 1 (Right) and Class 2 (Left).

Clear separation in several components indicates effective spatial filtering and discriminative power between motor-imagery classes.

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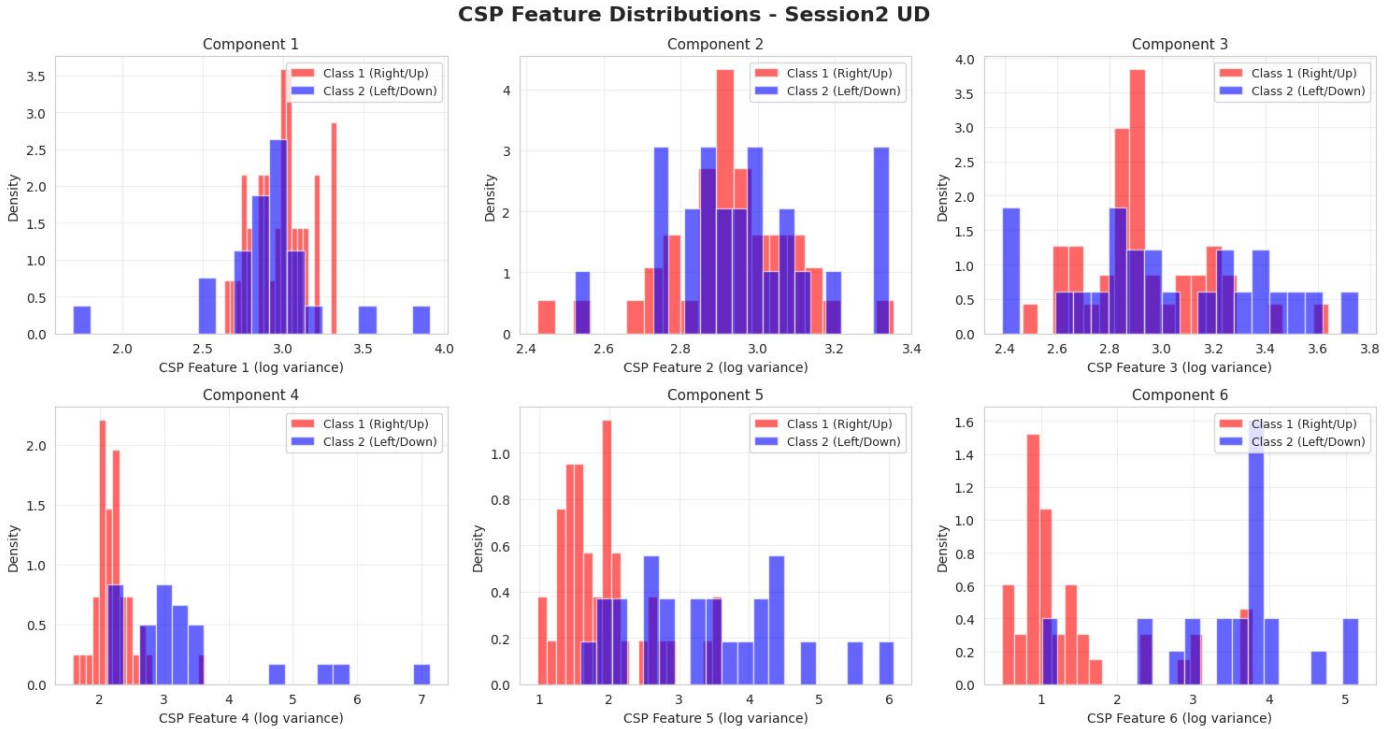
Subject 2 (LR Task)



The left panel shows training and test loss over 50 epochs, indicating stable convergence but some oscillation in test loss. The right panel shows accuracy curves, with training accuracy reaching 100% and test accuracy stabilizing around ~71%, suggesting mild overfitting but good generalization overall.

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Subject 2 (UD Task)



Histograms show the log-variance of six CSP components for Class 1 (Up) and Class 2 (Down).

Clear separation in several components indicates effective spatial filtering and discriminative power between motor-imagery classes.

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Subject 2 (UD Task)



The left panel shows training and test loss over 50 epochs, indicating stable convergence but some oscillation in test loss. The right panel shows accuracy curves, with training accuracy reaching ~97% and test accuracy stabilizing around ~89%, suggesting mild overfitting but good generalization overall.

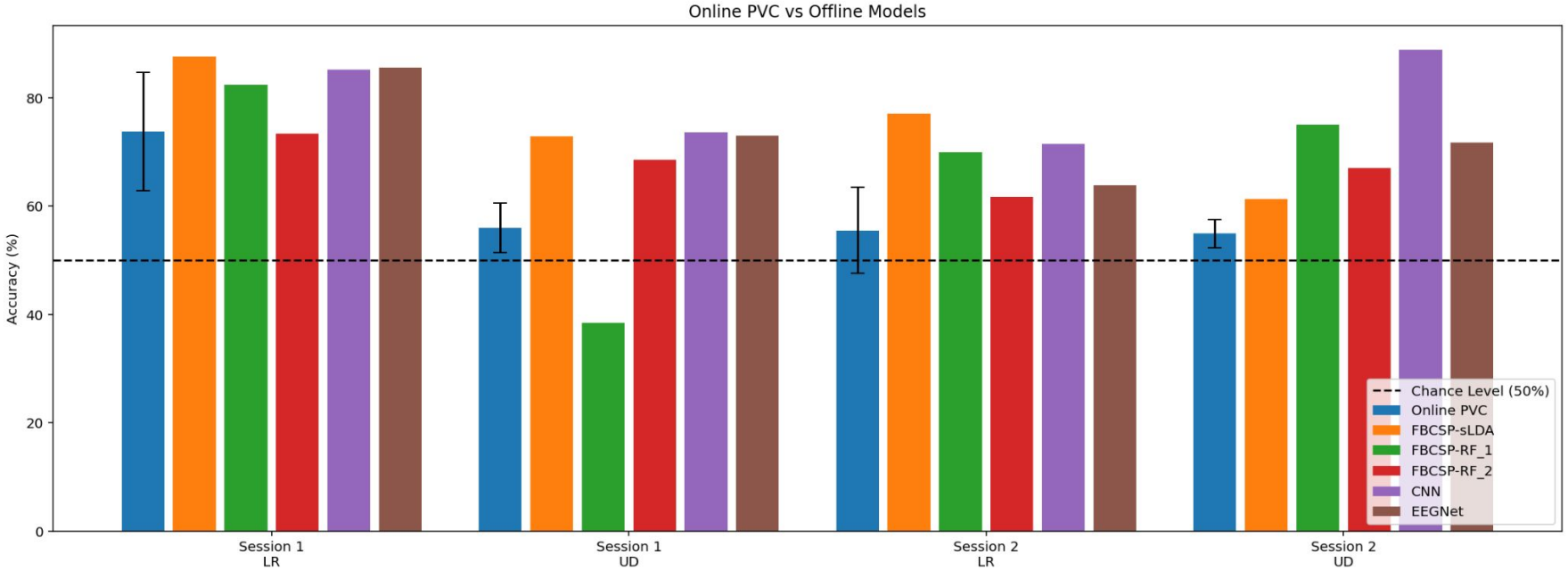
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Summary of Best Models

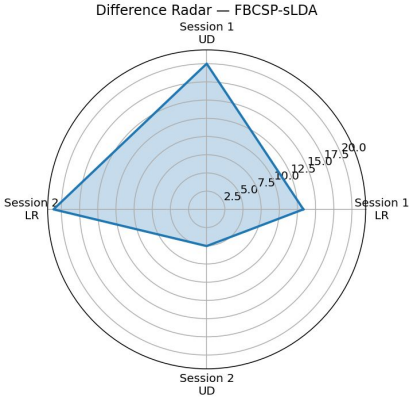
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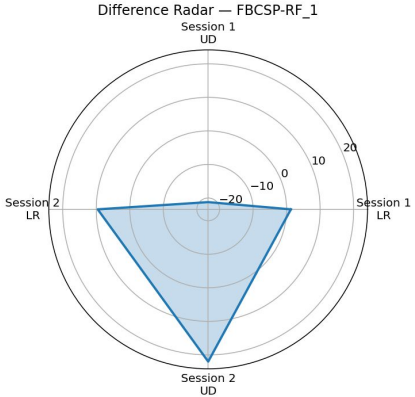
Final PVC Performance

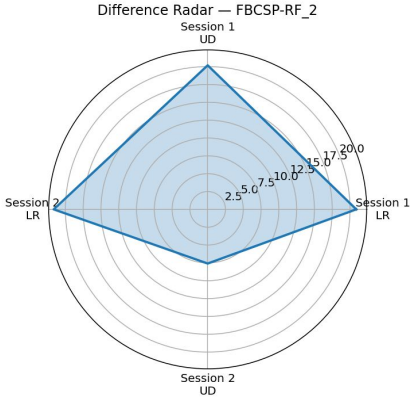


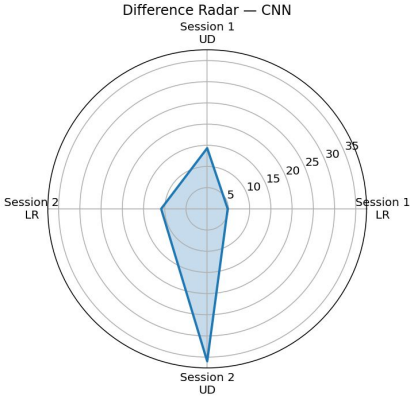
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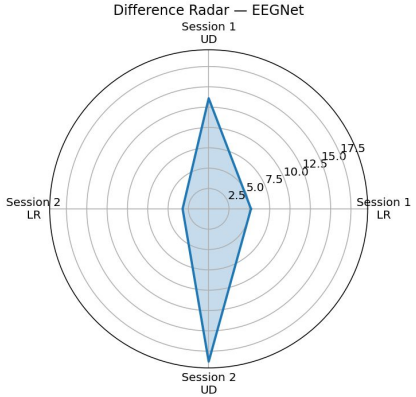
Difference Radar Between Online & Offline PVC





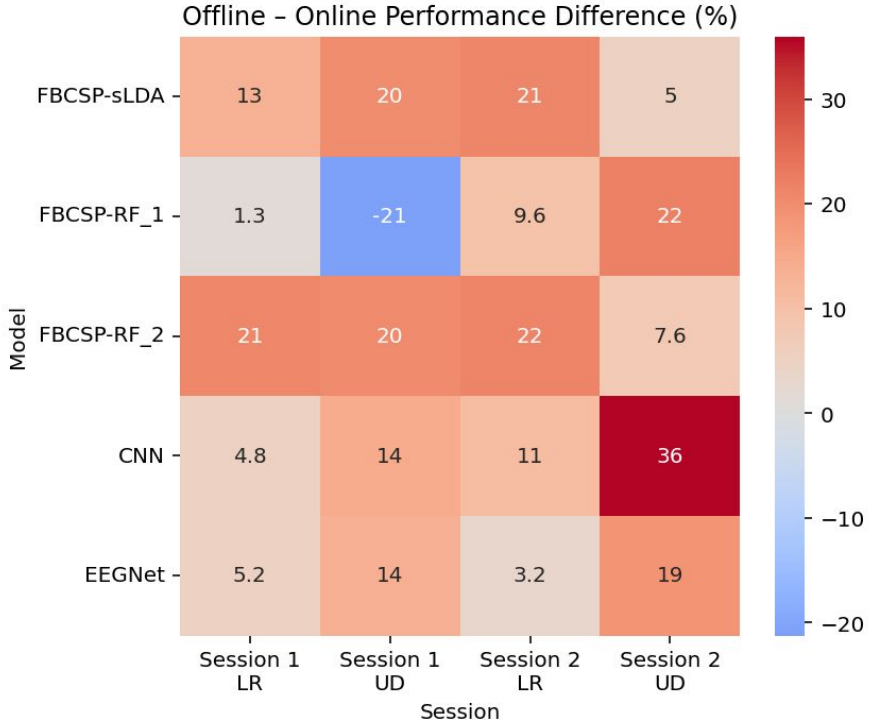






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Difference Radar Between Online & Offline PVC




```
Paired t-tests (offline vs online)

FBCSP-sLDA      t=4.019, p=0.02767
FBCSP-RF_1      t=0.318, p=0.77105
FBCSP-RF_2      t=5.260, p=0.01339
CNN             t=2.431, p=0.09322
EEGNet          t=2.801, p=0.06779
```

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Conclusion

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Conclusion

- In general, across all models**, our offline models outperformed the online classifier.**
  - Why? CSP/FBCSP can learn spatial patterns between multiple electrodes.
  - Deep learning performs better as they can learn non-linear relationships within the data that the online classifier and CSP classifiers cannot.

- **No single model won outright.** CSP/FBCSP + LDA models performed similar to our top performing models.
  - In the field of BCI, **preprocessing drives performance.**

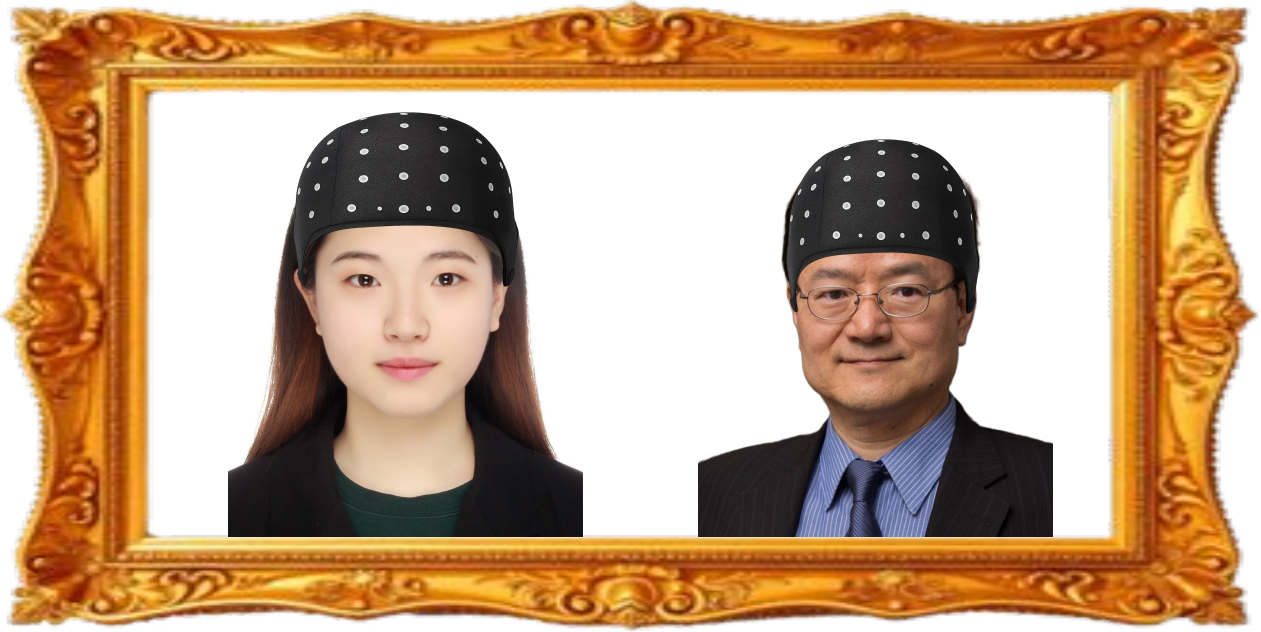
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Limitations and Future Work

- Data leakage
  - EEG is time-series, and we perform 5-fold cross validation, mixing in trials across time.
  - Proper validation requires having designated training and testing sections during experimentation.
  - However, we are limited by small dataset so only testing with one chronological split would likely lead to unstable PVC and overfitting.

- Future work
  - Implement online versions of these classifiers to address data leakage.
  - Test on more subjects to improve model generalizability.

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Thank you so much to Dr. He and Yidan for their help on this project!

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References

1. Lawhern, V. J., Solon, A. J., Waytowich, N. R., Gordon, S. M., Hung, C. P., & Lance, B. J. (2018). *EEGNet: A compact convolutional neural network for EEG-based brain–computer interfaces*. Journal of Neural Engineering, 15(5), 056013. https://doi.org/10.1088/1741-2552/aace8c
2. Kai Keng Ang, Zheng Yang Chin, Haihong Zhang and Cuntai Guan, "Filter Bank Common Spatial Pattern (FBCSP) in Brain-Computer Interface," *2008 IEEE International Joint Conference on Neural Networks (IEEE World Congress on Computational Intelligence)*, Hong Kong, China, 2008, pp. 2390-2397, doi: 10.1109/IJCNN.2008.4634130. keywords: {Classification algorithms;Filter bank;Filtering algorithms;Electroencephalography;Band pass filters;Gabor filters;Accuracy},

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